

Exploring the Role of Project Management in the Development of the Academic Agile Software Discourse: A Bibliometric Analysis

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ABSTRACT ■

The practical applications of agile methods and their impact on the productivity and efficiency of software development dominate the agile literature. We analyzed 827 academic articles with bibliometric techniques to explore the role project management research played in the development of the academic agile discourse. Bibliometric analyses over two time periods reveal that project management-related topics form a distinct stream of research in the second time period but not in the first. Furthermore, our results reveal that the academic agile discussion has been mainly unidirectional. This situation offers many opportunities for project management researchers to contribute to the agile discourse.

KEYWORDS: Scrum; agile software development; projects; co-citation analysis

INTRODUCTION ■

Since the publication of the *Agile Manifesto* (Agile Alliance, 2001) agile models have taken a central role in the practice of software development (West, Grant, Gerush, & D'Silva, 2010; Dingsøyr, Nerur, Balijepally, & Moe, 2012). Following the agile principles, many different agile-based process models have been developed to improve individual and/or organizational software development productivity (Dingsøyr, Dybå, & Moe, 2010).

Software development process models provide the conceptual scheme for managing the software development process by directing and coordinating software development activities and resources (Boehm, 1988). Project management and agile concepts share many conceptual similarities and, not surprisingly, agile software development (ASD) is already been mentioned in the general project management standards (Project Management Institute [PMI], 2013). Some authors even see specific agile software development models, such as Scrum, as a synonym for project management (Schwaber & Beedle, 2001).

Research has focused mainly on exploring the implementation, effectiveness, and efficiency of agile models (Lindvall et al., 2002; Li, Moe, & Dybå, 2010). It is unclear, however, whether and how project management methods and concepts could complement current agile methods. To better understand the potential merits, it is important to analyze the role project management has played in the development of the academic agile software development discourse.

In this study we identify the main intellectual structures of the academic agile discourse over time and explore how project management research was integrated into this discourse. We use bibliometric methods to reveal prominent research streams as well as their proximities and chronological relevance for the agile and project management communities. This method provides quantitative measures for evaluating the impact of project management research on the development of agile software development, their interrelations, and how they have evolved over time. The method has been successfully used to explore specific and more general questions in different management disciplines, such as innovation management (Teichert, 2004), operations management (Pilkington & Meredith, 2009), and project management (Artto, Martinsuo, Gemünden, & Murtoaro, 2009; Artto & Wikström, 2005; Hanisch & Wald, 2012). Some authors conducted comprehensive analyses of the agile discourse to identify main authors and research

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articles (Chuang, Luor, & Lu, 2014) and to identify general topics (Dingsøyr et al., 2012). However, the agile literature is dominated by practice-oriented publications; under these conditions, it is difficult to separate between practical and theoretical developments. Dingsøyr et al. (2012) offered important insights into the structure of the agile discourse at large but they have not focused on the changes of the academic agile discourse over time and the role of project management. Analyzing the existing academic agile literature and its relations to project management provide a comprehensive understanding of the central research themes and their links with each other. The revealed relations lead to promising questions for further research in both areas; specifically, we explore the following questions:

1. What were the dynamics in the intellectual structures of the academic agile discourse over time?
2. What role does project management play in the academic agile discourse?
3. What are evolving intellectual structures in the academic agile discourse?

The article is structured as follows. We briefly address the foundation of agile software development, followed by a discussion of the research methodology; then we present the results of our bibliometric study. The subsequent sections include the discussion of results followed by conclusions that address limitations and potential fruitful areas for future research.

The Foundation of Agile Software Development

The management of software development projects is dominated by the rules and principles of the *Agile Manifesto* (Dingsøyr et al., 2012). The idea for the *Agile Manifesto* evolved from the critique of the inflexibility of traditional software development process models and their negative effects on the outcomes of software development projects. The *Agile Manifesto* does not specify a specific

software development model, but rather defines a platform of rules and principles that allow the derivation of many models with different attributes. Over the past decades, many agile-based software development models have been introduced, including eXtreme Programming (XP), Scrum, lean software development, feature-driven development, and crystal methodologies, etc. (Dingsøyr et al., 2012).

Many researchers believe that agile-based process models overcome the often criticized shortcomings by providing greater flexibility, which leads to higher productivity (Cardozo, Neto, Barza, Franca, & da Silva, 2010) and efficiency (Li et al., 2010; Lindvall et al., 2002). Because of these effects, agile-based process models have quickly become the norm for software development. Unlike traditional linear plan-driven models, agile models are rooted in iterative and incremental steps by which software is developed through collaboration between self-managed, cross-functional teams (Conboy & Fitzgerald, 2004). Flexibility is a fundamental characteristic of the process models as expressed by the *Agile Manifesto*:

"Individuals and interactions over processes and tools. Working software over comprehensive documentation. Customer collaboration over contract negotiation. Responding to change over following a plan" (Agile Alliance, 2001, p. 1).

The resulting flexible, lightweight, and iterative processes facilitate changes and the ability to address a relatively unpredictable environment. As agile software development process models are becoming more and more popular among practitioners, academia is also showing increasing interest in this topic (Dingsøyr et al., 2012).

Methodology and Data

We used bibliometric co-citation analysis to quantitatively ascertain the structural patterns and research streams of the academic agile discourse (White & McCain, 1998; White, 1990; White & Griffith, 1981). It is based on the

underlying assumption that closely related works are frequently cited together (Culnan, 1986; Hoffman & Holbrook, 1993; Tahai & Meyer, 1999) and it is used to identify the most influential publications, authors, journals, and keywords, and to analyze relational links between them.

In this study, we employed the often used Document Co-citation Analysis (DCA) rather than Author Co-citation Analysis (ACA). Author co-citation analysis does not allow for the identification of central topics and their relationships (Teichert & Shehu, 2010), because many authors have contributed different thoughts and ideas to the agile discourse but they have also made contributions to other research fields. Using documents as the unit of analysis provides a more specific theme-related focus and precision about a research field other than the author-level analysis (Pilkington & Meredith, 2009; Teichert & Shehu, 2010).

The downside of using the document co-citation analysis method is that a great amount of effort is necessary to identify accurate articles. Because of the different journal and conference citation policies, or even misspellings, the database shows several entries for the same publication. All articles of the final selection of this study were carefully checked for these irregularities.

The basic procedure of extracting common interests and capturing main research streams within the body of citations is the principle component analysis (PCA). In this study, we applied principle component analysis with Varimax rotation to achieve simplified expressions of extracted factors. Multi-dimensional scaling (MDS) is commonly used to obtain graphical outputs (Ramos-Rodríguez & Ruíz-Navarro, 2004; Nerur, Rasheed, & Natarajan, 2008); however, it is only possible to graphically present a multi-dimensional scaling with two dimensions, which might not best fit the data (Culnan, 1986; Leydesdorff & Vaughan, 2006; Pilkington & Meredith, 2009).

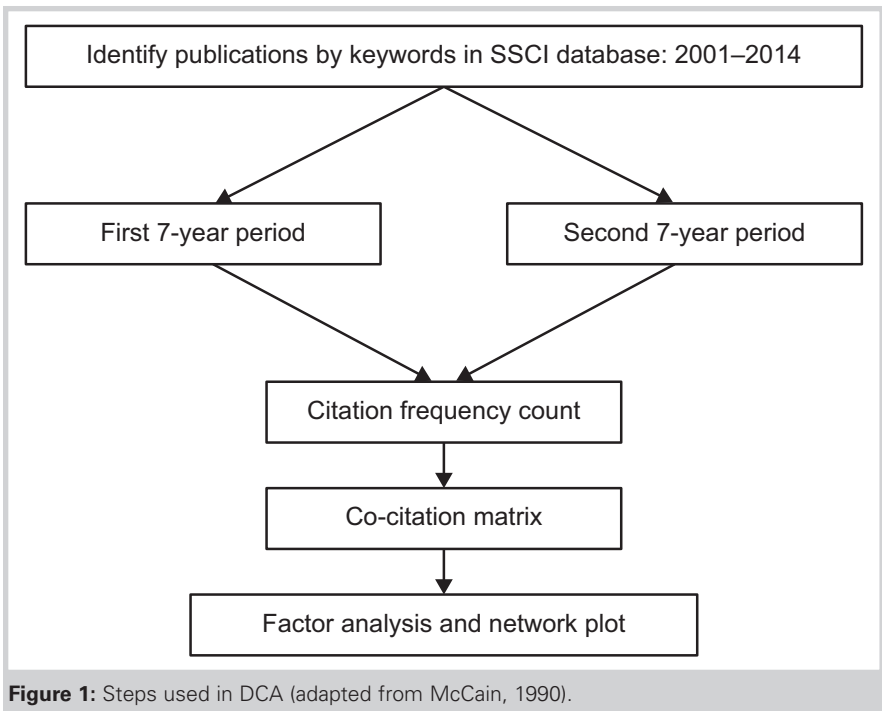


Figure 1: Steps used in DCA (adapted from McCain, 1990).

Many new visualization techniques have been developed to overcome this disadvantage (Wasserman & Faust, 1994). We applied UCINET (Borgatti, Everett, & Freeman, 2002), an analytical software, to plot the co-citation network. This method uses normalized multi-dimensional scaling data, but it is less sensitive to the high multidimensionality of the data (Pilkington & Meredith, 2009; Teichert & Shehu, 2010).

In our analysis we covered the timeframe from 2001 through 2014. We chose the year 2001 as our starting point because the *Agile Manifesto* was published in that year. To reveal changes in the research streams of the agile discourse we split the 14-year timeframe

into two time periods of seven years each (2001–2007 and 2008–2014). We did not choose shorter sub-periods because of the time delay between the initial publication and its citations. The seven-year range is sufficient to reveal basic structures in the discourse within the period, while also allowing for the observation of changes across two periods (Nerur, Mahapatra, & Mangalaraj, 2008). The steps we took for our study are described in Figure 1.

We chose the Social Science Citation Index (SSCI) as the database for identifying relevant publications because it contains a very broad set of scientific journals, books, conference proceedings, and other scientific publications.

Identify Publications from SSCI database and Bibliometric Descriptives (Step 1)

A search for keywords in the Social Science Citation Index database was used in order to identify agile publications. The keywords are names of specific agile models, agile project management, and their acronyms. We used project management as an explicit keyword to identify those publications that discuss project management-related topics in the context of agile and vice versa. Figure 2 illustrates the keywords and search logic.¹

“Agile” is not a specified term that only appears in the software development domain; it is also an important term in the astroparticle physics, production, supply chain, and some other fields. Therefore, we manually excluded those articles that were not related to software development or the project management domain. We also limited the research area to four categories: “computer science,” “business economics,” “operational research management science,” and “information science.” We only considered academic articles from journals and conferences because they better represent the academic discourse. Other authors, including Chuang et al. (2014) have identified the most influential authors and articles but their database included both academic and practitioner publications. Practitioner publications typically contain very few references and some do not have any reference lists and focus mainly on practical but not theoretical issues. Some journals and conference proceedings publish both scientific and practitioner articles, such as *Communications of the ACM*, *Agile Conference*; therefore, we paid special attention to articles from these types of sources.

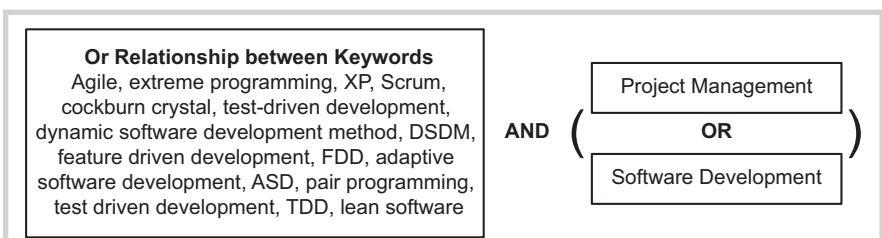


Figure 2: Search keywords and logic.

¹The search team is: TS = ((agile OR XP OR ("extreme programming") OR scrum OR ("cockburn crystal") OR dsdm OR ("dynamic software development method") OR ("feature driven development") OR fdd OR asd OR ("adaptive software development") OR ("pair programming") OR ("lean software") OR tdd OR ("test driven development") OR ("test-driven development")) AND ("software development" OR "project management"))

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Year	01	02	03	04	05	06	07	08	09	10	11	12	13	14	Total
Total Publication/Year	7	16	36	39	82	68	58	66	92	66	63	69	80	85	827
Scrum/Year	0	0	2	1	5	4	1	8	8	14	16	9	20	20	108
PM/Year	0	0	0	3	5	5	1	5	9	7	4	3	9	12	63

(PM = project management)

Table 1: Number of articles published per year from 827 selected articles.

Eventually, we retrieved 827 academic articles in this step. These articles were published by 394 different journals and conferences, and represented 690 different first authors. The number of academic agile publications per year grew rapidly between 2001 and 2005, and has been fairly stable since then, with an average of approximately 80 articles per year (Table 1). We identified a total of 63 project management-related articles and 108 Scrum-related articles from the retrieved 827 articles. The number of articles per year that address project management-related topics has been increasing since 2013. The frequencies of Scrum publications are also listed, because the concept of Scrum is viewed as conceptually close to project management. The number of Scrum-related articles has increased since 2008 and has been increasing per year accordingly. The most frequent authors and article sources are listed in Table 2. The authors listed have contributed ten or more academic articles of the 827 selected articles, and can be considered as main contributors to the development of the academic agile discourse. Information and software engineering journals as the top outlets are no surprise, because their main focus is

on research and experiences that contribute to the improvement of software development practices.

Citation Frequency Count (Step 2)

The data set we obtained from the previous step was split into two seven-year periods. The first seven-year period (2001–2007) contains 293 articles. The second seven-year period (2008–2014) contains 534 articles. A bibliometric software tool called BibExcel (Persson, Danell, & Schneider, 2009a, 2009b) was employed to compute citation and co-citation data. Since BibExcel only recognizes exactly coinciding terms, a manual normalization process is needed to guarantee accuracy. After processing data by BibExcel, the 293 identified articles from the first seven-year period resulted in approximately 1,390 references. The 534 identified articles from the second seven-year period resulted in approximately 3,564 references. Then we constructed the citation frequency list, which is a rank-ordered list showing the frequency that each document was cited by each sub-period data set.

Co-citation Matrix Compilation (Step 3)

In order to decide which documents to include in the document co-citation

analysis, two complimentary indicators were used: the Citations Received (CR) from each sub-period in our database, and the overall Times Cited (TC) as listed in the Social Science Citation Index database. First, the number of citations received indicates the relevance of a document for the agile discourse. Typically, the top 1% to 5% of publications from the citation frequency list is selected to conduct the analysis (Teichert, 2004; Small, 2006). In our study, we selected the top 5%. Second, the number of times cited from the Social Science Citation Index database indicates the relevance of the document to topics reaching beyond the agile discourse; therefore, an index was calculated for each sub-period by dividing citations received with times cited. Higher values indicate a stronger focus on agile, whereas lower values indicate a broader orientation of an article. We have not considered articles with an index number lower than 0.02 because they should have a very low impact on the agile discourse. In this step, therefore, we removed all articles with a low index from the co-citation analysis.

Eventually, we selected 31 documents for the first seven-year period and 102 documents for the second

Most Prolific Authors		Most Prolific Sources	
Names	# Articles	Article Sources (Journals and Conferences)	# Articles
Abrahamsson	22	<i>Information and Software Technology</i>	33
Maurer	14	<i>Journal of Systems and Software</i>	28
Dingsøyr	11	<i>Empirical Software Engineering</i>	16
Williams	10	<i>Agile Development Conference, Proceedings</i>	15
Moe	10	<i>Product-focused Software Process Improvement, Proceedings</i>	15

Table 2: The most prolific authors and article sources.

seven-year period and formed symmetrical co-citation matrices. The elements on the diagonal of the co-citation matrix were treated as missing values (McCain, 1990).

Factor Analyses and Network Analyses (Step 4)

First, we performed principle component factor analysis by using the two co-citation matrices as data input. Varimax rotation was used to achieve simplified expressions of subfields, which are helpful to better identify and interpret the extracted factors. Factors are often extracted based on the similarity of articles. The most popular articles are more likely to load on multiple factors and consequently do not necessarily drive the factors (Pilkington & Meredith, 2009). Horn's Parallel Procedure was employed to decide how many factors should be retained in our study (Horn, 1965; Glorfeld, 1995; Mertler & Vannatta, 2002). The underlying theory of Horn's Parallel Procedure is that factors from real data with a valid factor structure should have Eigenvalues greater than the 95th percentile of the Eigenvalues derived from random generated data with the same number of variables. First we generated a matrix of random data with the same size and the same number of variables as our co-citation matrix; then we ran the principle component analysis with our datasets and random

datasets. By comparing the Eigenvalues of our principle component analysis with the 95th percentile of the random data principle component analysis, we kept only factors with Eigenvalues greater than the random data.

Based on Horn's Parallel Procedure, we retrieved five factors for the first seven-year period, which account for 63% of the total variance, and nine factors in the second seven-year period, which also account for 63% of the total variance. Only loadings above ± 0.45 are reported, and only loadings greater than ± 0.7 are considered sufficient to interpret a certain factor (Nerur et al., 2008). Approximately 10% of the articles loaded on more than two factors.

Factor analytical methods seek solutions of maximal independence between factors and cannot provide any insights about how identified research streams are interrelated (Teichert, 2004). Therefore, we used the network plot function of UCINET to generate a map based on the publications' similarity in order to identify the interrelations between the research streams.

Results

Factor Analysis 2001–2007

The first seven-year period is represented by 31 publications, and the factor analysis resulted in five factors as documented in Table 3 below.

Factor 1-1 Test Driven Development (TDD): consists of eight co-cited articles (20.8% variance) discussing the test-driven development method. The test-driven method is critical for the analysis, design, and testing of software. The identified articles are primarily concerned with evaluating the test-driven method (Maximilien & Williams, 2003; Williams, Maximilien, & Vouk, 2003; Erdogmus, 2005) and comparing it with traditional software development methods (George & Williams, 2004).

Factor 1-2 Empirical Analysis: consists of six articles (15.8% variance) discussing the empirical exploration of agile software development methods in different contexts. The articles were published between 1998 and 2004 and they cover different topics, including software engineering (Fitzgerald, 1998; Kitchenham et al., 2002; Lindvall et al., 2002) the usage of XP (Abrahamsson & Koskela, 2004; Layman, Williams, & Cunningham, 2004) and the extension of agile methods (Dingsøyr & Hanssen, 2002).

Factor 1-3 XP: contains six co-cited articles (10.5% variance) that are mainly concerned with analyzing eXtreme Programming (XP). Erickson, Lyytinen, and Siau (2005) review the research analyzing XP and agile modeling; others explore practice with XP (Sharp & Robinson, 2004) and compared it with other software development methods (Mannaro, Melis, & Marchesi, 2004).

Factor 1-1: TDD	Factor 1-2: Empirical Analysis	Factor 1-3: XP	Factor 1-4: Agile Adoption	Factor 1-5: Method Engineering
George (2004)*	Abrahamsson (2004)*	Mannaro (2004)*	Cao (2004)*	Brinkkemper (1996)*
Erdogmus (2005)*	Fitzgerald (1998)*	Sharp (2004)*	Fitzgerald (2006)*	Henderson-Sellers (2005)
Maximilien (2003)*	Lindvall (2002)	Layman (2006)*	<i>Kitchenham (2002)</i>	
Williams (2003)*	<i>Kitchenham (2002)</i>	Erickson (2005)*		
Basili (1999)*	<i>Dingsøyr (2002)</i>	Turk (2005)		
<i>Dingsøyr (2002)</i>	<i>Layman (2004)</i>	<i>Layman (2004)</i>		
Boehm (1988)				
Hulkko (2005)				
% explained 20.8	15.8	10.5	8.8	7
* = Loadings above 0.7	No * = Loadings between 0.45 and 0.70	<i>Italic</i> = Cross-loading		

Table 3: Factor analysis for the period 2001–2007.

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Factor 1-4 Agile Adoption: consists of three articles (8.9% variance) that discuss the issues and merits of agile adoption in different contexts (Cao, Mohan, Xu, & Ramesh, 2004; Fitzgerald, Hartnett, & Conboy, 2006).

Factor 1-5 Method Engineering: includes two articles (7.0% variance) discussing method engineering, which is a defined process that is used to create new or to improve existing software development methods (Brinkkemper, 1996). Method engineering provides greater flexibility to agile and can also support the improvement of existing software development processes (Henderson-Sellers & Serour, 2005).

Factor Analysis 2008–2014

The second seven-year period is represented by 102 publications and the factor analysis resulted in nine factors as documented in Table 4.

Factor 2-1 Agile Adoption: is comprised of 36 articles (30.6% variance). The articles discuss the adoption of agile specific methods, evaluate their benefits, and conceptually discuss team-related assumptions of agile or agile-based methods as well as related topics of system development.

One group of articles discusses the challenges and merits (Pikkariainen, Haikara, Salo, Abrahamsson, & Still, 2008; Petersen & Wohlin, 2009) of agile adoption in different contexts (Cao, Mohan, Xu, & Ramesh, 2009; Ilieva, Ivanov, & Stefanova, 2004; Mannaro et al., 2004; Fitzgerald et al., 2006; Karlström & Runeson, 2006; Layman, Williams, & Cunningham, 2006; Salo & Abrahamsson, 2008; Vijayasathy & Turk, 2008; Chan & Thong, 2009; Mangalaraj, Mahapatra, & Nerur, 2009; Misra, Kumar, & Kumar, 2009) and the introduction of an adoption framework for agile (Qumer & Henderson-Sellers, 2008). The empirical results indicate that the appreciation for agile software development practices increases after their implementation.

Many agile methods are team based; thus, not surprisingly, agile team-related

topics are analyzed by a sub-group of articles (Faraj & Sproull, 2000; Marks, Mathieu, & Zaccaro, 2001; Salas, Sims, & Burke, 2005; Moe, Dingsøyr, & Dybå, 2010; Kirkman & Rosen, 1999; Hoda, Noble, & Marshall, 2012). Another sub-group of articles analyzes software development through the lens of systems engineering by integrating agile with systems development frameworks (Orlikowski, 1993; Iivari & Huisman, 2001; Paetsch, Eberlein, & Maurer, 2003; Bjørnson & Dingsøyr, 2008; Vidgen & Wang, 2009).

Factor 2-2 Agile Project Management: is the second strongest factor that contains 25 articles (7.6% variance) that mainly discuss agile project management (APM) and also the relationship between agile and project management standards and processes.

The management challenges of software development projects with agile methods or principles are analyzed from practical (Sutherland, Viktorov, Blount, & Puntikov, 2007; Korkala & Abrahamsson, 2007; Leybourne, 2009) and theoretical (Turk, France, & Rumpe, 2005; Maruping, Venkatesh, & Agarwal, 2009; Dybå & Dingsøyr, 2008) perspectives.

Another sub-group of articles analyzes the integration of agile methods with existing traditional project management standards and processes (Abrahamsson, Warsta, Siponen, & Ronkainen, 2003; Coram & Bohner, 2005; McAvoy & Butler, 2009; Harris, Collins, & Hevner, 2009; Karlström et al., 2006).

Factor 2-3 Test-Driven Development (TDD): represents nine articles (5.4% variance) that analyze test-driven software development. Examples of test-driven development related topics include the advantages and impact on software quality (Madeyski, 2005; Canfora, Cimitile, Garcia, Piattini, & Visaggio, 2006; Janzen & Saiedian, 2006; Nagappan, Maximilien, Bhat, & Williams, 2008). Other articles discuss the practice of test-driven development (Maximilien et al., 2003; Sanchez, Williams, & Maximilien, 2007) and test driven-related experiments designed to

explore the effectiveness of test driven practices (George & Williams, 2004) and influences (Müller & Höfer, 2007).

Factor 2-4 Agile and Plan-Driven Method: contains seven articles (4.1% variance) critically comparing specific agile software development with traditional plan-driven software development (Erickson et al., 2005). Some articles focus on advantages that agile software development has instead over plan-driven software development (Petersen & Wohlin, 2010; Laanti, Salo, & Abrahamsson, 2011; Chow & Cao, 2008).

Factor 2-5 Agile in Global Context (Global Software Development): contains four articles (3.7% variance) discussing the managerial challenges for agile-based software development in global settings (Sarker & Sarker, 2009; Holmström, Fitzgerald, Ågerfalk, & Conchúir, 2006). Other topics include communications in a global software development environment (Herbsleb & Mockus, 2003), and the use of Scrum in globally distributed software development teams (Paasivaara, Durasiewicz, & Lassenius, 2009).

Factor 2-6 Pair Programming: is comprised of four articles (3.4% variance) discussing the concept of pair programming. The articles analyze the impact of pair programming on software development productivity and software quality (Müller, 2005; Lui & Chan, 2006; Madeyski, 2006; Begel & Nagappan, 2008).

Factor 2-7: Method Engineering: includes two articles (3.0% of variance) discussing method engineering. The factor is defined by the same articles as those in the first period (Brinkkemper, 1996; Henderson-Sellers & Serour, 2005).

Factor 2-8 Scrum: contains seven articles (2.7% variance) discussing the effects of Scrum on the software development process and outcomes in general (Baskerville, Pries-Heje, & Madsen, 2010), and, more specifically, the achieved overall customer satisfaction (Mann & Maurer, 2005) and its application in the context of distributed teams

Factor 2-1 Agile Adoption	Factor 2-2 Agile Project Management	Factor 2-3 TDD	Factor 2-4 Agile and Plan-driven	Factor 2-5 Agile In Global Context	Factor 2-6 Pair Programming	Factor 2-7 Method Engineering	Factor 2-8 Scrum	Factor 2-9 XP
Staples (2007)* Qumer (2008)* Pikkariainen (2008)* Salo (2008)* Moe (2010)* Cao (2009)* Hoda (2012)* Kirkman (1999)* Misra (2009)* Chan (2009)* Vijayasathya (2008)* Ferreira (2007)* Orlikowski (1993)* Paetsch (2003)* Mangalaraj (2009)* Faraj (2000)* Salas (2005) Marks (2001)	Fitzgerald (2006) Takeuchi (1986) Moore (1991) Ilieva (2004) Fitzgerald (1998) Björnson (2008) Greer (2004) Mannaro (2004) Layman (2006) Karlström (2006) Petersen (2009) Layman (2004) Vidgen (2009) Schwaber (1997) Paasivaara (2009) Erickson (2005) Hikka (2005) Iivari (2001) Korkala (2007) Sharp (2009) Turk (2005) Hossain (2009) Karlström (2006) Vidgen (2009) Sharp (2008) Dingsøyr (2008) Boehm (1988)	Sanchez (2007)* Müller (2007)* George (2004)* Janzen (2006)* Maximilien (2003) Nagappan (2008) Müller (2001) Madeyski (2005) Canfora (2006)	Petersen (2010)* Chau (2003) Laanti (2011) Baskerville (2004) Chow (2008) Dingsøyr (2012) Erickson (2005)	Sarker (2009)* Holmström (2006)* Herpsleb (2003)* Paasivaara (2009)	Müller (2005)* Lui (2006)* Begel (2008)* Madeyski (2006)*	Brinkkemper (1996)* Henderson— Sellers B., (2005)	Mann (2005) Baskerville (2010) Conboy (2009) McAvoy (2009) Paasivaara (2009) Sutherland (2007) Harri (2009)	Bahli (2005) Cao (2004) Sharp (2008) Hikka (2005) Turk (2005)
% explained	30.6	5.4	4.1	3.7	3.4	3	2.7	2.6
* = Loadings above 0.7 No* = Loading between 0.45–0.69 <i>Italic</i> = Cross loading								

Table 4: Factor analysis for the period 2008–2014.

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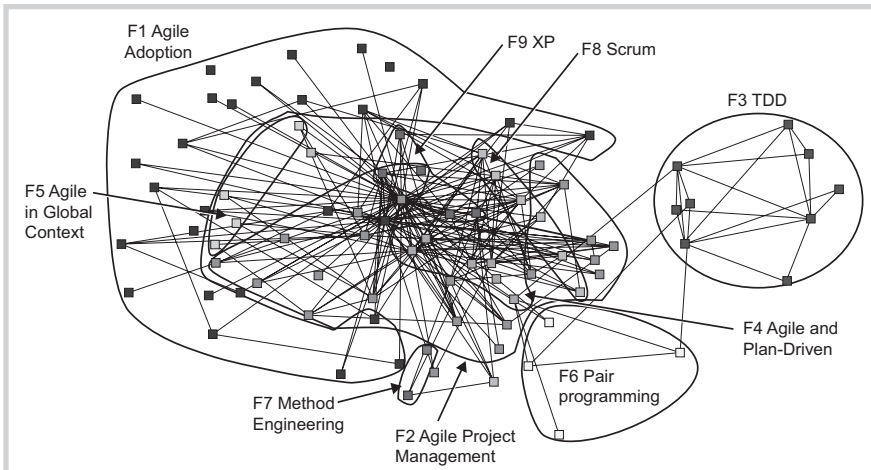


Figure 3: Network plot for the period 2008–2014.

(Sutherland et al., 2007; Paasivaara et al., 2009). Furthermore, some articles explore conceptually the context of Scrum (Conboy, 2009) and Scrum in the agile context in broader terms (McAvoy et al., 2009; Harris et al., 2009).

Factor 2-9 XP: is comprised of five articles (2.6% variance) that mainly analyze the effects of eXtreme Programming (XP) on knowledge creation (Bahli & Zeid, 2005), team collaboration and coordination (Sharp & Robinson, 2008), and its innovativeness (Hilkka, Tuure, & Rossi, 2005). Other articles analyze the agile software development process in more general terms (Turk et al., 2005).

Network Plots

In this step we use a network map to visualize the identified factors of the document co-citation analysis and their relationships with each other. A close proximity of publications on the map usually suggests that there is considerable commonality between these publications. The first seven-year period network analysis presents a relatively loose network structure, which is related to the adolescent status of the agile discourse within the observed time period and the relative low number of publications. The network plot is not presented here because the relations between the clusters are not offering new insights.

Figure 3 presents the network plot for the second seven-year period. For ease of presentation, the co-citation threshold was set to three, which means that only articles and their linkages are displayed when they are co-cited three times or more (Pilkington et al., 2009; Teichert et al., 2010). The plot shows that some topics are “closer” in distance to each other than to others. Test-driven Development and Pair Programming, for example, are somewhat isolated self-contained topics that are positioned on the periphery; both methods are used to improve software productivity on the individual developer level and the factors therefore represent discussions of specific communities that are more isolated from the general agile discourse. This also applies to Factor 2-7 representing the method engineering discussion as mentioned in sub-period one. The proximities of the other factors such as Agile Adoption (Factor 2-1), Agile Project Management (Factor 2-2),

Agile and Plan-Driven (Factor 2-4), and Scrum (Factor 2-8) are more general topics and define the core of the academic agile discourse.

Discussion

Exploring the Dynamic Development of the Academic Agile Discourse

We analyzed the academic agile discourse over two sub-periods to explore its time-dependent dynamics. A comparison of the identified intellectual structures between the first and the second seven-year periods demonstrates: (1) that some topics from the previous period continue to be important and were even further developed; (2) that one topic faded (Empirical Analysis); and (3) that several new topics emerged (see Table 5).

In general, the discovered intellectual structures for the two periods are consistent with other analyses of the agile discourse (Dingsøyr et al., 2012); however, our time-based analysis reveals some interesting dynamics in the intellectual structure. All agile-specific topics such as agile adoption, XP, test-driven development, and method engineering that we identified in the first period remain important for the academic discourse in the second period. Articles related to these topics might not be the same between the two periods because some were highly cited in the first period but received low citations in the second period and vice versa. Only the topic of agile-related empirical studies “disappeared,” which is not surprising because this topic is broad and covers several aspects of agile. The differentiation of topics

Disappeared Topics	Continued Topics	Emerged Topics
• Empirical Analysis	• Agile Adoption • TDD • Method Engineering • XP	• Agile Project Management • Agile and Plan-driven • Agile in Global Context • Pair Programming • Scrum

Table 5: Development of the agile discourse over time

began later with the increase of academic research as reflected by the new topics of the second period.

Our co-citation analysis for the first seven-year period (2001–2007) resulted in five factors. The number of articles constituting these factors varies and is in some cases relatively low; however, all of the articles are consistently co-cited, suggesting a generally accepted intellectual structure. The topics that we uncovered explore the improvement of software productivity at the developer level (XP and test-driven development), the introduction and application of agile-related methods and process models within organizations (agile adoption and empirical analysis of agile), and engineering perspectives on developing and modifying software methods (method engineering). For the first period we could not identify a distinct discussion of project management-related topics. Within that time period we could identify several practitioner-based publications discussing agile project management (Runeson, 2005; Augustine, Payne, Sencindiver, & Woodcock, 2005; Ceschi, Sillitti, Succi, & Panfilis, 2005) but a distinct topic in the academic discourse integrating project management-related concepts into agile software development did not evolve.

The second seven-year period (2008–2014) is characterized by a strong growth of agile-related academic research that is reflected in more than 100 co-cited articles. The co-citation analysis revealed nine distinct factors. The main focus of the discourse remains on the introduction and implementation of new agile models (test-driven development, XP, and Scrum). We also identified a distinct discourse of project management-related topics (see Factor 2-2). Of the nine factors, it is also the second strongest and takes a central role as it viewed in the network plot (see Figure 3). Other topics are more practice oriented, such as the use of agile methods in the global software development context (Holmström et al., 2006).

The Role of Project Management in the Academic Agile Discourse

Project management research is becoming increasingly relevant to the agile discourse. Despite the fact that several authors refer to the management of software projects as agile project management (Augustine et al., 2005; Sutherland et al., 2007; Fernandez & Fernandez, 2008), it is surprising that we were not able to identify a distinct topic during the first time period. This clearly changed for the second period in which we identified 25 academic articles exploring the interaction between agile software development and project management concepts. These articles largely analyze the management challenges using agile methods (Sutherland et al., 2007; Korkala & Abrahamsson, 2007; Leybourne, 2009; Turk et al., 2005; Maruping et al., 2009; Dybå & Dingsøyr, 2008), and the integration of agile methods with existing traditional project management standards and processes (Abrahamsson et al., 2003; Coram & Bohner, 2005; McAvoy et al., 2009; Harris et al., 2009; Karlström et al., 2006). Another group of publications analyzes the conceptual differences between agile and plan-driven methods (Abrahamsson et al., 2003; Boehm, 1988; Dybå et al., 2008; Sharp et al., 2008).

Boehm and Turner (2004) discuss the problems involved in integrating agile and plan-driven methods and provide a “hybrid model” to balance agility with existing software project management standards.

A specific analysis of agile project management publications reveals that the most frequently cited articles (Table 6) were primarily published in software and computer science journals; only one on the list was published in a management journal. All articles discuss consequences of agile for traditional software project management.

In sum, the agile discourse largely focuses on integrating agile principles into the management of software development projects (Nerur, Mahapatra, & Mangalaraj, 2005; Leybourne, 2009). The reverse direction is rarely considered; in other words, project management models have solely served as a subject of change by agile methods, yet project management research has not influenced the development and analysis of agile methods.

Evolving Intellectual Structures in the Academic Agile Discourse

The analysis of the agile-based academic literature also indicated some specific

First Author	Source	Title	TC	CR
Sutherland (2007)	<i>HICSS 2007</i>	Distributed scrum: agile project management with outsourced development teams	190	22
Coram (2005)	12th IEEE International Conference and Workshops ECBS'05	The impact of agile methods on software project management	112	17
Fernandez (2008)	<i>Journal of Computer Information Systems</i>	Agile project management—agilism versus traditional approaches	83	16
Leybourne (2009)	<i>International Journal of Managing Projects in Business</i>	Improvisation and agile project management: a comparative consideration	29	16
McAvoy (2009)	<i>European Journal of Information Systems</i>	The role of project management in ineffective decision making within agile software development projects	45	10

TC = Times Cited, CR = Citations Received

Table 6: The top five most referenced academic agile project management publications.

Exploring the Role of Project Management

themes that are not yet visible in the intellectual structures of the academic agile discourse but have the potential to evolve as distinct topics within the coming years. Agile in the global context of software development, with particular emphasis on distributed team settings, seems to be an evolving topic in the academic agile discourse (see also Dingsøyr et al., 2012). One specific theme is related to the communication and coordination required to overcome “distance” in globally distributed teams using XP (Holmström et al., 2006; Layman, Williams, Damian, & Bures, 2006; Korkala et al., 2007), including the coordination and control of distributed teams (Kirsch, 1997; Maruping et al., 2009; Harris et al., 2009), and the use of other agile methods in global software development contexts (Hossain, Babar, & Paik, 2009; Fruhling & Vreede, 2006; Kirsch, 2004; Ramesh, Cao, Mohan, & Xu, 2006). We expect that research in the global software context will continue to grow and will likely represent a distinct topic in the agile discourse in the near future.

Several authors have already suggested and analyzed the concept of “hybrid” models by integrating agile principles into traditional software project management. Furthermore, agile principles have recently been applied in other management-related research fields. The integration of agile principles in stage-gate models (Cooper, 2016; Cooper & Sommer, 2016) and in other management models of NPD development (Conforto & Amaral, 2010; Fekri, Aliahmadi, & Fathian, 2009) are currently discussed. The discussions on “hybrid” models strongly suggest that they will form a distinct topic within the academic agile discourse.

Conclusions

The goals of this article were to discover distinct research streams in the agile movement, to address the changes in these streams over time, and clarify the role that project management-related themes played in the development of

the academic agile discourse. We chose a bibliometric analysis to quantitatively analyze the academic agile literature. A thorough search in the Social Science Citation Index database identified 827 agile-related publications. A limitation of this database is that it does not list all types of publications, for example, books and outlets other than journals and conferences. This limitation did not directly affect our results because our research goal was to analyze the academic agile discourse; therefore, we focused on academic publications. Many practice-oriented books and articles were cited in the identified academic publications; these practice-oriented publications are important for the general understanding, use and dissemination of methods and models but they focus on practical problems and often do not address any specific research question or theoretical development. Another limitation is the relatively adolescent status of the agile discourse, which is reflected in the dynamics between the two time periods analyzed; it is very likely that some of the research streams that were identified might disappear or that others might evolve over the next few years. However, software development-related research topics are aging very quickly, thus it seems that a 14-year time span should be sufficiently long to identify some fundamental intellectual structures in the discourse.

The motivation for our bibliometric analysis was to explore the impact of project management research on the agile software development discourse. For this analysis we deliberately chose a quantitative approach to gain a broad understanding of the state of the agile discourse and the role that project management takes. The majority of topics that we identified are related to specific agile methods, including test-driven development, eXtreme programming, pair programming, and Scrum. Furthermore, we could identify some conceptual discussions, such as the conceptual analysis of agile versus plan-driven approaches and method engineering.

Our results demonstrate that the management of projects is increasingly recognized in the agile software development discourse, and that the management aspect has grown to become an important distinct topic. We have identified several research articles that compare or integrate agile project management with project management practices. This trend will continue, and agile principles are expected to play an important role in the future discourse of project management. Despite their affinity, it is not clear which role project management research has played in the development of the academic agile discourse; thus far, agile-related research is largely published within the outlets of the software development domain. In contrast, only few articles have been published in management research journals covering topics such as the adoption of agile in different contexts other than software development (Conforto, Salum, Amaral, da Silva, & de Almeida, 2014; Sommer, Hedegaard, Dukovska-Popovska, & Steger-Jensen, 2015). Several authors in the project management field are discussing how agile methods could be scaled to large projects (Petit & Besner, 2013; Dingsøyr, Fægri, & Itkonen, 2014; Hobbs & Petit, 2016).

However, the integration of agile topics into the project management discourse is mainly unidirectional. Classic topics in the project management discourse are currently underrepresented or have not yet found their way into the agile discourse. This situation provides an opportunity to analyze the central concepts of project management within the context of agile.

The management of risk and its different concepts and tools is a key process in the project management standards (Akintoye & MacLeod, 1997; Raz & Michael, 2001; Ward & Chapman, 2003), which is also relevant for agile software development but has yet to be picked up by the agile literature. Some other central project management topics could be interesting for the further conceptual

development of agile: linking agile with the multi-project management and multi-tasking environment (Ebert & De Neve, 2001; De Maio, Verganti, & Corso, 1994); project governance (Abednego & Ogunlana, 2006), project stakeholder management (Littau, Jujagiri, & Adlbrecht, 2010; Achterkamp & Vos, 2008); knowledge management and creation in the project environment (Bresnen, Edelman, Newell, Scarbrough, & Swan, 2003; Fong, 2003); linking strategy to project and portfolio implementation (Meskendahl, 2010; De Reyck et al., 2005), internal IT development versus external integration of contractors (Sun & Meng, 2009); and change control management (Kemerer, 1995; Wang, Ju, Jiang, & Klein, 2008).

Our study suggests the need for more conceptual research and the development of a theoretical perspective on agile. These areas of research will facilitate the conceptual integration of project management-related concepts into the agile discourse but might also influence the theoretical discourse in project management at large. Project management research offers interesting research questions for the agile discourse. This direction poses an attractive opportunity for project management researchers to publish agile-related research in management-oriented research outlets.

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